

Hands-on Tasks:

(A) Malware Behavior Analysis

Use Cuckoo Sandbox or Hybrid Analysis to analyze a sample malware (use test files, NOT real malware).

Identify:

File system changes.

Network activity.

Registry modifications.

(B) Digital Forensics Task

Install Autopsy (or FTK Imager) and analyze a forensic disk image.

Extract and report evidence (deleted files, browsing history, logs).

Submission:- document everything and push it as week 3 Day 2

(A) Analysis of Malware Behavior

1. Configuration and Implementation

Cuckoo Sandbox was installed on Kali Linux.

Cuckoo was launched, and the web interface was accessed at `http://localhost:8000`.

EICAR test file (`eicar.com`) was used as a malware sample for testing.

The file was submitted to Cuckoo Sandbox for analysis.

2. Results:

File System Changes: - Because it was a test file, no questionable file changes were found.

- Although the file was identified as test malware, it did not carry out any destructive actions.

Network Activity:- No outgoing connections were made by the test file.

Registry Modifications:- No registry modifications were found, which is pertinent to malware that runs on Windows.

(B) Task for Digital Forensics :

1. Configuration and Implementation

Autopsy was installed on Kali Linux by running `sudo apt install autopsy -y`.

I launched Autopsy and went to `http://localhost:9999`.

A forensic disk image in the.E01 format was loaded for examination.

2. Results

Recovery of Deleted Files:

found and restored a number of deleted files from the disk image.

exported a few chosen files for additional review.

Analysis of Browsing History:

retrieved browsing history from the profiles of Chrome and Firefox.

discovered downloaded files, stored cookies, and visited URLs.

Examination of System Logs:

System event logs (.evtx files for Windows, /var/log for Linux) were examined.

discovered reports of system errors and login activity.